The Two Borel–Cantelli Lemmas

An End-to-End Meta-Math Walkthrough

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Abstract

The summable case bounds the entire tail union. The independent divergent case instead bounds the probability that an entire long tail block contains no occurrence. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

PS-01 / Probability and Stochastic Processes / Report v1

I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For events A_n on a probability space, write $\{A_n \text{ i.o.}\} = \bigcap_{N\geq 1} \bigcup_{n\geq N} A_n$. Prove: (i) if $\sum_n \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 0$; (ii) if the events are mutually independent and $\sum_n \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$. Mutual independence means every finite subcollection has product probability, including the corresponding complements.*

A. Foundations and permitted imports

Use countable subadditivity, continuity of probability along monotone event sequences, finite-product independence, and $1 - p \leq e^{-p}$ for $0 \leq p \leq 1$. These follow respectively from countable additivity and elementary real analysis. The first direction has no independence assumption.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

No inter-node proof imports are needed: the nodes establish separate directions under the common theorem contract.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Summable tail control

Object and role. Bound a tail union and let the summable tail vanish. **Local obligation:** Under summability, $\mathbb{P}(A_n \text{ i.o.}) = 0$.

Inputs. $\sum_n \mathbb{P}(A_n) < \infty$ and $B_N = \bigcup_{n \geq N} A_n$.

C — Contract

Publish E1/v1 only after establishing this obligation: Under summability, $\mathbb{P}(A_n \text{ i.o.}) = 0$.

K — Knowledge

$\sum_n \mathbb{P}(A_n) < \infty$ and $B_N = \bigcup_{n \geq N} A_n$. Retain the stated assumptions and import versions.

S — Strategy

Bound a tail union and let the summable tail vanish.

V — Validation

No products or independence are used in this direction.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Countable subadditivity gives $\mathbb{P}(B_N) \leq \sum_{n \geq N} \mathbb{P}(A_n)$. The right side tends to zero because the series converges.

N1.2 The events B_N decrease and their intersection is the infinitely-often event. Continuity from above yields its probability as $\lim_N \mathbb{P}(B_N) = 0$.

Export E1/v1. Under summability, $\mathbb{P}(A_n \text{ i.o.}) = 0$.

Acceptance gate. No products or independence are used in this direction.

Targeted repair. If independence is imposed in this branch, remove it from the local inputs: the union bound already supplies the full conclusion. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Independent divergent tail control

Object and role. Estimate the complementary event of no occurrence in a finite tail block and then exhaust the tail. **Local obligation:** Under independence and divergence, $\mathbb{P}(A_n \text{ i.o.}) = 1$.

Inputs. Mutual independence and $\sum_n \mathbb{P}(A_n) = \infty$.

C — Contract

Publish E2/v1 only after establishing this obligation: Under independence and divergence, $\mathbb{P}(A_n \text{ i.o.}) = 1$.

K — Knowledge

Mutual independence and $\sum_n \mathbb{P}(A_n) = \infty$. Retain the stated assumptions and import versions.

S — Strategy

Estimate the complementary event of no occurrence in a finite tail block and then exhaust the tail.

V — Validation

Divergence of the series alone does not give the finite product identity; mutual independence is used there.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Fix N. For $m \geq N$, independence gives $\mathbb{P}(\bigcap_{n=N}^{m} A_n^c) = \prod_{n=N}^{m}(1 - \mathbb{P}(A_n)) \leq \exp(-\sum_{n=N}^{m} \mathbb{P}(A_n))$. Divergence makes this tend to zero as $m \to \infty$.

N2.2 By continuity from above, $\mathbb{P}(\bigcap_{n \geq N} A_n^c) = 0$, so $\mathbb{P}(B_N) = 1$. The intersection of the countably many probability-one events B_N still has probability one, proving the second assertion.

Export E2/v1. Under independence and divergence, $\mathbb{P}(A_n \text{ i.o.}) = 1$.

Acceptance gate. Divergence of the series alone does not give the finite product identity; mutual independence is used there.

Targeted repair. If the product step is used for dependent events, withhold this export and either verify independence or use a separately justified dependence condition. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Summable tail control

1. **N1.1.** Countable subadditivity gives $\mathbb{P}(B_N) \leq \sum_{n \geq N} \mathbb{P}(A_n)$. The right side tends to zero because the series converges.
2. **N1.2.** The events B_N decrease and their intersection is the infinitely-often event. Continuity from above yields its probability as $\lim_N \mathbb{P}(B_N) = 0$.

B. N2 — Independent divergent tail control

1. **N2.1.** Fix N. For $m \geq N$, independence gives $\mathbb{P}(\bigcap_{n=N}^{m} A_n^c) = \prod_{n=N}^{m}(1 - \mathbb{P}(A_n)) \leq \exp(-\sum_{n=N}^{m} \mathbb{P}(A_n))$. Divergence makes this tend to zero as $m \to \infty$.
2. **N2.2.** By continuity from above, $\mathbb{P}(\bigcap_{n \geq N} A_n^c) = 0$, so $\mathbb{P}(B_N) = 1$. The intersection of the countably many probability-one events B_N still has probability one, proving the second assertion.

□

V. WORKED ORCHESTRATION

For independent fair coin tosses, let A_n be the event that toss n is heads. The sum of probabilities diverges, so heads occur infinitely often almost surely. For a separate sequence of events B_n with $\mathbb{P}(B_n) = 2^{-n}$, the sum is finite, so only finitely many B_n occur almost surely regardless of their dependence.

A. Boundary and hypothesis audit

Let $A_n = A$ for every n, where $\mathbb{P}(A) = 1/2$. Then $\sum_n \mathbb{P}(A_n) = \infty$, but the infinitely-often event is just A, of probability $1/2$. This disproves the divergent implication without a dependence hypothesis.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 inference: a divergent sum of probabilities implies probability one of infinitely many occurrences without independence. **Validation witness:** identical events

$A_n = A$. **Diagnosis:** Strategy replaced the probability of an intersection by a product without authorization. **Repair:** use mutual independence only in the second branch, as in §1. The first branch remains valid for arbitrary dependence; its export need not be revised.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Apply the summable half to error events in PS-02; the probability bound must be summable over all sample sizes.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Amir Dembo, Probability Theory: STAT310/MATH230 notes for measure-theoretic probability and limit arguments; Stanford STAT310C syllabus and lecture materials for martingale inequalities and stopping theorems. The versions and proofs used in this collection are stated explicitly.